ANNA-SETU: PROJECT BRIEF & PHASED TIMELINE

1M1B AI for Sustainability Virtual Internship • In Collaboration with IBM SkillsBuild & AICTE

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Primary SDG: SDG 12 (Target 12.3 & 12.5)	**Dual-Track Model:** Track A (AI App) + Track B (Hardware)	**Milestones:** Prototype: End-Nov 2026 | Deploy: End-Dec 2026

1. Executive Project Brief, Problem & SDG Alignment

Anna-Setu (from Sanskrit: *'Anna'* meaning sustenance and *'Setu'* meaning bridge) is a closed-loop socio-technical framework designed for institutional campus dining ecosystems. It attacks food waste at two interconnected points: **1) At the kitchen level**, where unforecasted safety margins create 10%–30% surplus buffers, and **2) At the diner plate level**, where students routinely take excessive portions onto their plates and dump uneaten leftovers into garbage bins.

By coupling an AI predictive software engine (**Track A**) with an on-site smart plate-waste interceptor and accelerated bio-recycling converter (**Track B**), Anna-Setu creates a zero-waste circular campus economy: untouched kitchen food is safely redistributed to verified night shelters, while plate scraps are converted on-site into high-grade organic bio-fertilizer, relieving municipal authorities (Lucknow Nagar Nigam) and preventing fugitive methane emissions in landfills.

Table 1.1: Sustainable Development Goal (SDG) Alignment Matrix

UN SDG	Specific Target	Anna-Setu Implementation & Impact
SDG 12: Responsible Consumption *(Primary)*	**Target 12.3:** Halve food waste by 2030. **Target 12.5:** Substantially reduce waste via recycling.	**Track A** reduces kitchen overcooking by 15–25% via predictive ML; **Track B** deters over-serving with behavioral plate nudges and upcycles plate waste into organic fertilizer.
SDG 2: Zero Hunger *(Secondary)*	**Target 2.1 & 2.2:** End hunger; ensure access to safe, nutritious food.	Untouched surplus is checked through FSSAI statutory holding algorithms and safely dispatched to local night shelters (Rain Basera Charbagh, KGMU, Gomti Nagar).
SDG 11: Sustainable Cities	**Target 11.6:** Reduce urban municipal per-capita environmental impact.	Eliminates organic wet-waste burden on municipal collection trucks and avoids ■5,000–■25,000 penalties under Lucknow Nagar Nigam Solid Waste By-Laws.
SDG 13: Climate Action	**Target 13.3:** Climate mitigation and institutional awareness.	Diverts organic matter from anaerobic landfill decomposition, eliminating methane (CH4) emissions. Real-time displays show diners their carbon and water footprints.

Target Beneficiaries: 1) Students: Dynamic nudges build mindful eating habits; **2) Mess Management:** Cuts procurement bills by 15–25%; **3) Lucknow Nagar Nigam:** Zero waste burden hauled to landfills; **4) Local Shelters:** Reliable access to verified nutritious meals; **5) Campus Grounds:** Free high-grade bio-fertilizer.

2. Dual-Track Integrated Solution Architecture

Anna-Setu operates two synchronized operational tracks bridging the physical dining floor with digital cloud intelligence:

TRACK A: ANNA-SETU AI SOFTWARE PLATFORM
(Status: In Progress / Working Prototype Deployed)

• **Module 1 (Dining Demand Advisor):** Time-series forecasting model predicting next-day cooking volumes based on turnstiles, exams, and weather.
• **Module 2 (RAG Compliance Engine):** Grounded in **FSSAI Surplus Regulations (2019)** (danger zone 5°C–65°C, 2h/4h ambient rule) and **Lucknow Nagar Nigam By-Laws** (45m transit buffer across 8 zones).
• **IBM Granite Model Compiler:** Generates WhatsApp/SMS alerts (<60w) and official **FSSAI Form B & LNN Gate Pass manifests**.
• **Live Assets:** Deployed on Hugging Face Spaces (Gradio) and local Streamlit app with 5 automated test cases.

TRACK B: SMART PLATE-WASTE BIO-RECYCLER
(Status: In Ideation / Hardware Prototype in Progress)

• **Behavioral Nudge Front-End:** Precision plate scale (0–20kg) at tray return counter + 7-inch display showing wasted grams, water footprint, and hostel leaderboards to stop diners from taking extra food.
• **Pre-Processing Shredder:** Reduces uneaten food scrap size to 3–5mm.
• **Accelerated Bio-Composting Chamber:** Thermophilic heating (55°C–65°C) + active aerobic aeration + microbial enzymes achieve **85%–90% volume reduction in 12–24h**.
• **Recyclable Resource Output:** High-grade **Organic Bio-Fertilizer / Soil Conditioner** for campus greenery and local farmers.
• **IoT Cloud Bridge:** ESP32 streams daily plate waste logs back to Track A AI.

Table 2.1: Track B Hardware Bill of Materials (BOM) & Cost Feasibility

Subsystem	Component	Technical Role in Station	Est. Cost (INR)
Compute / IoT	ESP32 WROOM-32 MCU	Wi-Fi telemetry, sensor polling, cloud API bridge	■450
Plate Waste Scale	0–20 kg Load Cell + HX711	Measures individual student plate waste left on tray	■550
Nudge Display	7-inch IPS / OLED Screen	Displays wasted grams, water footprint & leaderboard	■850
Thermal Sensing	MLX90614 Contactless IR	Verifies kitchen surplus food is held safe >=65°C	■750
Aerobic Aeration	12V Brushless Fan + Carbon	Supplies oxygen to composting chamber; stops odors	■250
Thermophilic Heat	100W PTC Ceramic Strip	Maintains 55°C–65°C to accelerate microbial breakdown	■450
Power & Housing	12V SMPS + SS Kiosk	Regulated power and IP65 sanitary stainless housing	■1,370
TOTAL BOM	**Complete Integrated Kiosk**	**Affordable, student-built IoT bio-recycler prototype**	**■4,670 (~$55)**

3. Project Timeline & Phased Execution Roadmap

The project follows a structured two-month Agile delivery plan anchored to program milestones: **Milestone 1 (End of November 2026):** Working prototypes of Track A and Track B integrated, with campus administration approvals; **Milestone 2 (End of December 2026):** Live on-campus 2-week pilot deployment, AI model retraining on real waste data, and final impact handover.

Phase / Sprint	Date Range	Key Work Deliverables & Activities	Gate Status
Phase 1 & 2: RAG Engine & UI	Nov 1 – Nov 10, 2026	Codified FSSAI 2019 and LNN by-laws into JSON KB; built TF-IDF vector retriever, compliance calculator, and IBM Granite notification compiler. Deployed Gradio demo on Hugging Face and local Streamlit app. Verified 5 automated test cases.	**COMPLETED**
Sprint 3A: HW Breadboard & IoT	Nov 11 – Nov 20, 2026	Procure ESP32, load cell, MLX90614 IR sensor, and display; assemble bench breadboard; flash C++/MicroPython firmware (esp32_smart_kiosk.ino); bridge telemetry to api_server.py.	**IN PROGRESS**
Sprint 3B: Approvals & Gate 1	Nov 21 – Nov 30, 2026	Draft permission letter to Dean of Student Welfare / Mess Committee (UPES); establish coordination agreement with Rain Basera Charbagh; **Milestone 1 Gate Review with 1M1B mentor**.	**MILESTONE 1 GATE** (End-Nov 2026)
Phase 4: Enclosure & Pilot	Dec 1 – Dec 21, 2026	Assemble IP65 stainless enclosure at tray return; conduct 14-day live campus pilot in student mess; log plate-waste reduction via nudge screen; dispatch compliant kitchen surplus; produce first batch of bio-fertilizer.	PLANNED PILOT (Central Mess)
Phase 5: Impact & Closeout	Dec 22 – Dec 31, 2026	Retrain Module 1 demand model on real pilot plate-waste logs; compute total kg rescued, kg bio-fertilizer harvested, methane avoided, and procurement ■ saved. Submit final PDF report, PPT, and demo video.	**MILESTONE 2 GATE** (End-Dec 2026)

Summary Note: The project tracker and activity task breakdown are maintained dynamically in the collaborative Google Sheets register: **PankhuriGovila_YIRP**, shared with **paramjeet@onemoneb.org**. The Excel source file is available at PankhuriGovila_YIRP_Project_Tracker.xlsx.